

All-Digital Background Calibration of a Successive Approximation ADC using the “Split ADC” Architecture

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Abstract—The “Split ADC” architecture enables fully digital calibration and correction of nonlinearity errors due to capacitor mismatch in a Successive Approximation (SAR) ADC. The die area of a single ADC design is split into two independent halves, each converting the same input signal. Total area and power is unchanged, resulting in minimal increase in analog complexity. For each conversion, the half-sized ADCs generate two independent outputs which are digitally corrected using estimates of capacitor mismatch errors for each ADC. The ADC outputs are averaged to produce the ADC output code. The difference of the two outputs is used in a background calibration algorithm which estimates the error in the correction parameters. Any nonzero difference drives an LMS feedback loop toward zero difference which can only occur when the average error in each correction parameter is zero. Simulation of a 16 bit 1Msps SAR ADC show calibration convergence within 200 000 samples.

Index Terms: Analog-digital conversion, adaptive systems, calibration, self-calibrating, digital background calibration, mixed analog-digital integrated circuits.

I. INTRODUCTION

THE successive approximation (SAR) architecture continues to be of interest for Nyquist rate analog-to-digital converters (ADCs) in nanoscale CMOS [1]. In particular, the charge-balance SAR ADC [2] is well suited to CMOS integration and offers a good compromise in the tradeoffs among sampling rate, power, dynamic range, and die area. For the charge-balance SAR, ADC linearity is limited by capacitor matching. As CMOS devices enter the nanoscale region, difficulty with matching in the presence of increased variability becomes more critical [3], necessitating some form of calibration. Since scaling of CMOS device dimensions offers clear advantages for digital circuitry in terms of density, speed, and integration, it becomes advantageous to push calibration into the digital domain if possible.

This paper presents an application of the “Split ADC” [4], [5] architecture to the problem of calibrating and correcting linearity errors in successive approximation ADCs. Compared with the work in [5], the novel content in this paper is the extension of the split ADC concept to the SAR architecture.

This paper is organized as follows: Section II reviews issues associated with errors in successive approximation ADCs, as well as analog and digital methods of correcting errors. Section III describes the novel aspects of the hardware design of the split SAR ADC. Section IV describes the error correction and calibration process using the split ADC architecture. Section V describes the design of a 16b 1Msps SAR ADC in a 180nm

CMOS process, with simulation results at the behavioral and extracted silicon levels.

II. SAR ADC REVIEW

Figure 1 shows a simplified version of charge-balance SAR ADC operation [6]. For simplicity only half of a differential implementation is shown; all results are applicable to the fully differential case. Also for simplicity, we will temporarily ignore offset error; the effect of offset will be discussed in Section IV-B. In the sample phase shown in Fig. 1a, the input signal v_{IN} is sampled on the bottom plates of all capacitors. The top plate voltage V_{TP} is sampled to V_{CM} , the desired common-mode voltage at the comparator input. Assuming for simplicity $V_{CM} = 0$ (V_{CM} cancels in a fully differential implementation), the charge on each capacitor is given by

$$Q_i = C_i v_{IN} \quad (1)$$

and the total charge Q_T on all capacitors is

$$Q_T = \sum_{i=1}^n C_i v_{IN} = C_T v_{IN} \quad \text{with} \quad C_T = \sum_{i=1}^n C_i \quad (2)$$

with C_T the total capacitance.

In the successive approximation process shown in figure 1b, the capacitor top plates are applied to the input of the comparator and the bottom plates are switched to reference voltages $\pm V_{REF}$ in accordance with the comparator decisions $D_k = \pm 1$, where D_k is the decision in the k th approximation cycle which is stored in the successive approximation register (S.A.R.). Since the comparator input ideally draws no current, the total charge on the capacitor top plates will remain constant. At the conclusion of the successive approximation process, the total charge on the capacitors is given by

$$Q_T = \sum_{i=1}^n C_i (D_i V_{REF} - v_{TP}) \quad (3)$$

where v_{TP} is the voltage at the comparator input. By charge conservation we equate (3) and (2) to obtain

$$C_T v_{IN} = \sum_{i=1}^n C_i (D_i V_{REF} - v_{TP}) \quad (4)$$

Rearranging (4) and solving for v_{IN}/V_{REF} gives

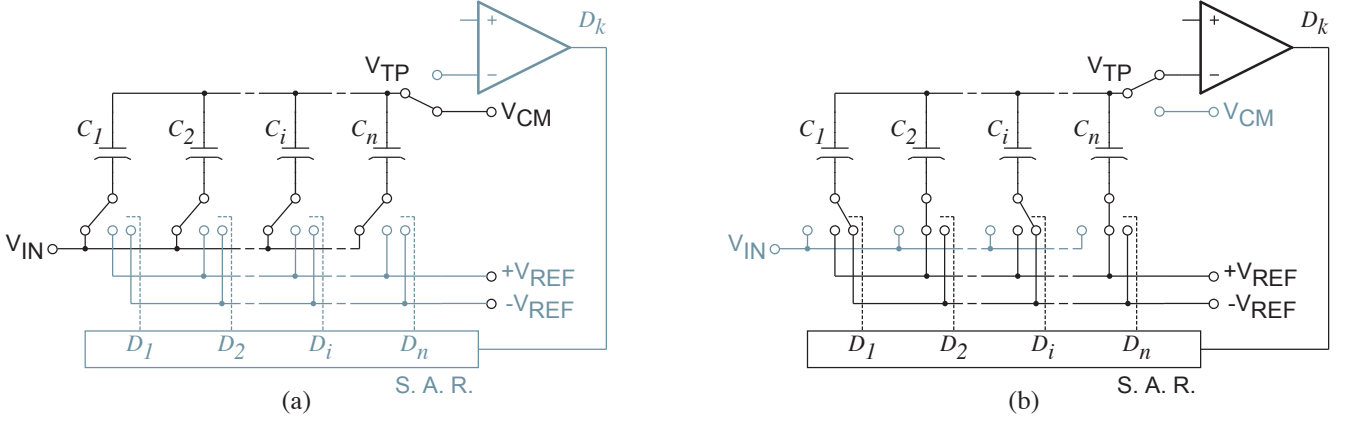


Fig. 1. SAR ADC Operation (a) Input Sampling (b) Completion of SAR process.

$$\frac{v_{IN}}{V_{REF}} = \sum_{i=1}^n \left(\frac{C_i}{C_T} \right) D_i - \frac{v_{TP}}{V_{REF}} \sum_{i=1}^n \left(\frac{C_i}{C_T} \right) D_i \quad (5)$$

The successive approximation process restores voltage v_{TP} to within an LSB of its original value; the second term in (5) therefore represents the quantization error. The first term in (5) is used to determine the output code of the ADC, a digital representation x of v_{IN}/V_{REF} by

$$x = \sum_{i=1}^n \underbrace{\left(\frac{C_i}{C_T} \right)}_{W_i} D_i \quad (6)$$

From (6) we see that the contribution of each decision D_i to the digital output x can be represented by a numerical “weight” W_i given by the fraction of each capacitor C_i relative to the total C_T . In the ideal case, the capacitor values are chosen so that the W_i form a binary weighted sequence $1/2^i$ and the decisions D_i are the bits in a binary representation of the output code x .

In a practical implementation, there are many considerations that make the design more complex than indicated by Figure 1. Usually a combination of segmentation, dummy structure, and common centroid layout techniques are used to improve capacitor matching [6]. To avoid an impractically large range of capacitor values in higher resolution ADCs, the simple parallel configuration of Figure 1 is often modified to include series elements in the configuration of the sampling and charge balancing capacitor network [2], [8]. To avoid large nonlinearity errors due to missing codes or missing decision levels, it may be necessary to provide additional capacitors and decisions for redundancy [8]. In all cases, however, a charge balance relationship equivalent to (4) will hold so that the relationship between decisions and the output code can be described by (6). To the extent that capacitor mismatch causes the weights W_i in (6) to deviate from the ideal $1/2^i$, some form of calibration and/or correction will be necessary to avoid ADC nonlinearity in the output code x .

There are several approaches to dealing with the mismatch problem. One approach is to simply size the capacitors large enough so that expected statistical deviation meets the mismatch limit for the desired ADC resolution. However, as

described in [3], this can impose severe die area and power dissipation penalties since the required capacitor size is often much greater than the limit set by kT/C noise.

Correction of ADC errors can be realized by analog or digital techniques, or some combination. An example of a combined analog and digital approach is given in [8], in which a table of calibration coefficients drives a DAC and injects a correcting charge into the charge balance node for each decision D_i . The coefficients, determined in one-time calibration procedure performed in the manufacturing process, restore the correct binary weighting in (6).

An example of a more digital approach is given in [7], in which a known training input is applied and an adaptive loop determines the correct W_i for linear ADC operation. When the unknown input signal is applied, the adaptive loop is “frozen” and the W_i values are used for correction of the ADC output codes. A disadvantage of this approach is that the known input signal is required; calibration cannot take place while the input signal is being processed.

The following sections describe the application of the Split-ADC approach to digital correction of ADC linearity for the SAR converter. By exploiting the advantages of CMOS scaling and performing all calibration and correction in the digital domain, requirements on analog circuitry are relaxed so that degraded accuracy can be tolerated. The calibration algorithm operates continuously in the background to estimate the decision weights W_i necessary in (6) for digital correction of the ADC output code x . The calibration algorithm places no undue requirements on input signal behavior, and the speed of calibration adaptation is sufficient to track parameter variations due to environmental change.

III. SPLIT SAR ADC

The block diagram of the Split SAR ADC system is shown in Figure 2. The analog portion of each SAR ADC is split into two independent halves, each comprising a charge-balance capacitive DAC, associated sampling and V_{REF} switch network, and comparator. On the digital side of Figure 2, correction is implemented using lookup tables (L.U.T.s) with decision weights W_{iA} and W_{iB} that are continuously updated in the background by an LMS digital calibration loop controlled by the error estimation block. The digital calibration and

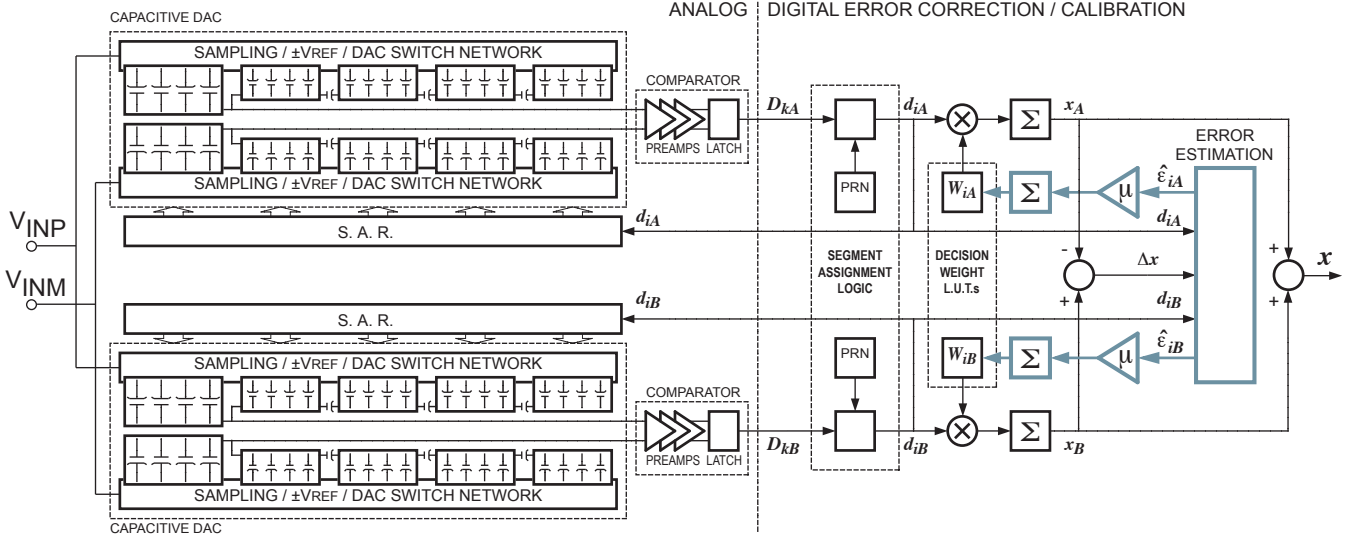


Fig. 2. Split SAR ADC Block Diagram.

correction are discussed in more detail in Section IV; the remainder of this section describes the analog circuitry.

A. Capacitive DAC

In the example implementation shown in Figure 2, the capacitive DAC is divided into 5 blocks each representing 4 comparator decision bits. A conceptual block diagram of the capacitive DAC implemented in this design is shown in Fig. 3. The total capacitance was determined from kT/C noise considerations. Bits 1-4 use 500fF unit capacitors; all remaining bits use 63fF unit capacitors. The design shown here makes heavy use of redundancy, with redundant bits included for every fourth decision. This is excessively conservative and was done to ease layout and ensure that eventual silicon could be calibrated, rather than to optimize area or SAR decisions. With the redundancy provided by overlap between the blocks, 16-b resolution is achieved with a total of 20 comparator decisions $D_1 - D_{20}$.

The following subsection describes a key novel feature of the hardware design: assignment of comparator decisions to capacitor segments is not fixed, but rather can be controlled on a conversion-by-conversion basis.

B. Dynamic Segment Assignment

Figure 4 illustrates the capacitor segment assignment process using the four MSBs of the converter as an example. Figure 4a shows the standard SAR function, with binary weighted capacitors $C_{b1} - C_{b4}$ switched to the $\pm V_{REF}$ reference as determined by the comparator decisions $D_1 - D_4$ loaded into the SAR. To improve matching, the capacitor values are realized using a unit capacitor array. Usually the assignment of segment capacitor switches to comparator decisions is fixed, often in a common centroid pattern which minimizes mismatch errors. In this work, however, the segment assignment logic (shown in Figure 4b) dynamically assigns the comparator decisions $D_1 - D_4$ to different segments. The decisions seen

by each segment's V_{REF} switch are denoted by the segment decisions $d_1 - d_f$ in Figure 4b.

Operation of the segment assignment logic is shown in conceptual form in Figure 4b. As shown in the diagram, groups of capacitor segments are assigned to each comparator decision as directed by a "base" pointer, which is varied in pseudorandom (PR) fashion on a conversion-by-conversion basis. In the example shown, the "base" pointer value is 5, so the first 8 segments beginning at location 5 ($C_5 - C_c$) represent comparator decision D_1 . The segment assignment logic routes comparator decision D_1 to segment decisions $d_5 - d_c$ which control the switches corresponding to capacitor segments $C_5 - C_c$. The switches for the next 4 segments, $C_d - C_0$, are controlled by comparator decision D_2 . Similarly, capacitors C_1 and C_2 are controlled by comparator decision D_3 and capacitor C_3 is controlled by comparator decision D_4 . Note that there will always be one segment which is not used; this ensures that as the "base" pointer is changed there will always be different segments used even in the case of a DC input.

In summary, the key novel feature of the hardware design is the calibration algorithm's use of a segment assignment logic block and the individually controlled segment switches to dynamically change the assignment of comparator bit decisions D_k to capacitor segment decisions d_i . This feature is key to the calibration algorithm described in section IV.

C. Matrix Representation

The operation of the segment assignment logic can be represented in matrix form; for the example of Figure 4b, the assignment of comparator decisions D_k to segment decisions d_i is given by (7). The segment assignment matrix $\mathbf{S}$ indicates how comparator decisions in $\mathbf{D}$ are assigned to each segment switch control in $\mathbf{d}$.

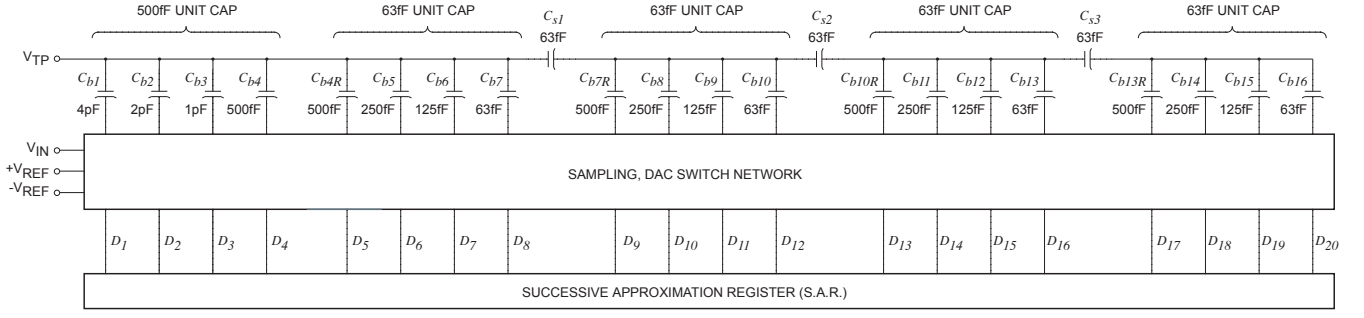


Fig. 3. Conceptual diagram of capacitive DAC.

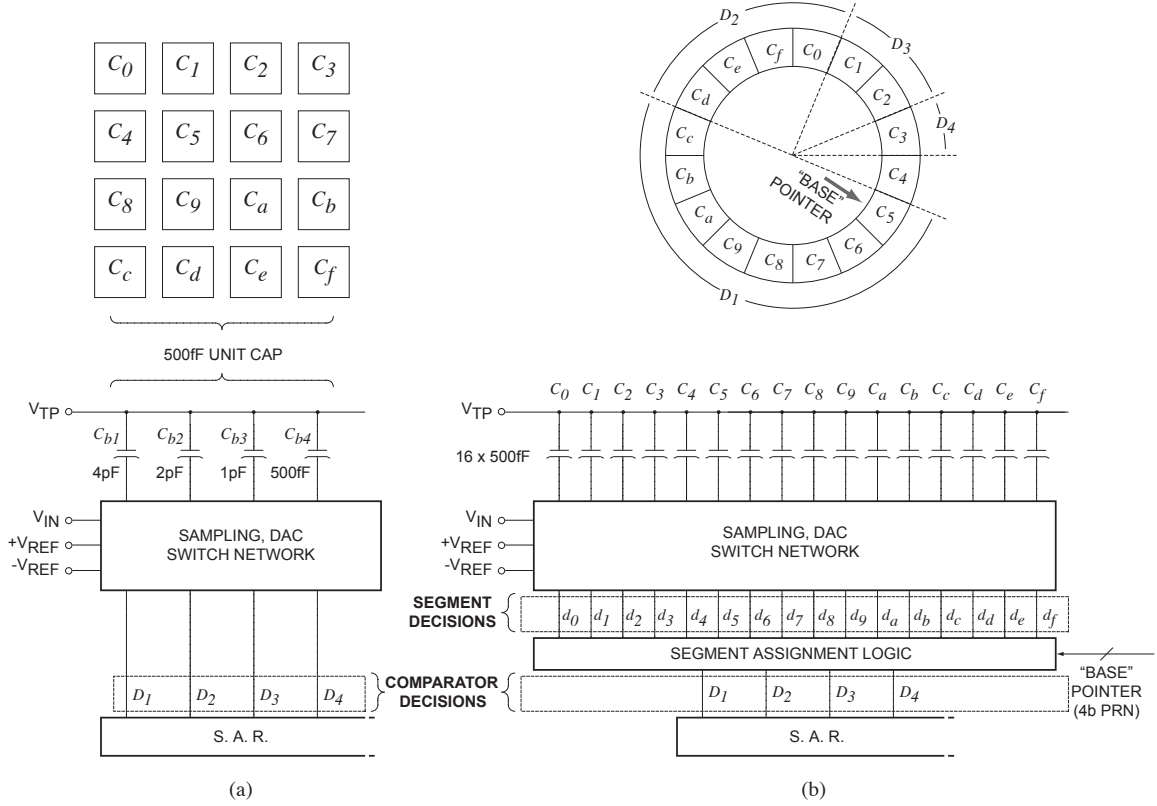


Fig. 4. (a) Conceptual arrangement of DAC capacitors for C_{b1} through C_{b4} representing bits b1-b4, and conceptual layout organization of 16 unit capacitors used to make up C_{b1} through C_{b4} (b) Actual arrangement of 16 unit capacitors, with conceptual view showing how assignment of unit capacitors to bit decisions can be varied conversion-to-conversion as controlled by a 4-bit pseudorandom "base" pointer.

$$\begin{array}{c} \text{d} \\ \left[\begin{array}{c} d_0 \\ d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \\ d_7 \\ d_8 \\ d_9 \\ d_A \\ d_B \\ d_C \\ d_D \\ d_E \\ d_F \end{array} \right] = \begin{array}{c} \text{S} \\ \left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right] \end{array} \begin{array}{c} \text{D} \\ \left[\begin{array}{c} D_1 \\ D_2 \\ D_3 \\ D_4 \end{array} \right] \end{array} \end{array} \quad (7)$$

IV. SPLIT-ADC CALIBRATION

A flow chart for the algorithm implemented by the digital correction and calibration block is shown in Fig. 5. The figure is partitioned into those calculations performed for correction in every conversion (right side), and those performed in the background to update calibration parameters (left side). The shaded boxes indicate the calculations implemented and the relevant equation numbers from the analysis that follows. For each of the "A" and "B" ADCs, the comparator decisions D_{kA} and D_{kB} are independently reassigned in pseudorandom (PR) fashion to segment decisions d_{iA} and d_{iB} as shown in Fig. 2. The corrected digital outputs x_A and x_B are accumulated from the segment decisions using a calibration LUT containing the segment decision weights W_{iA} and W_{iB} , which are continuously updated in the background.

The background calibration process, indicated on the left side of Fig. 5, begins with the x values populating an es-

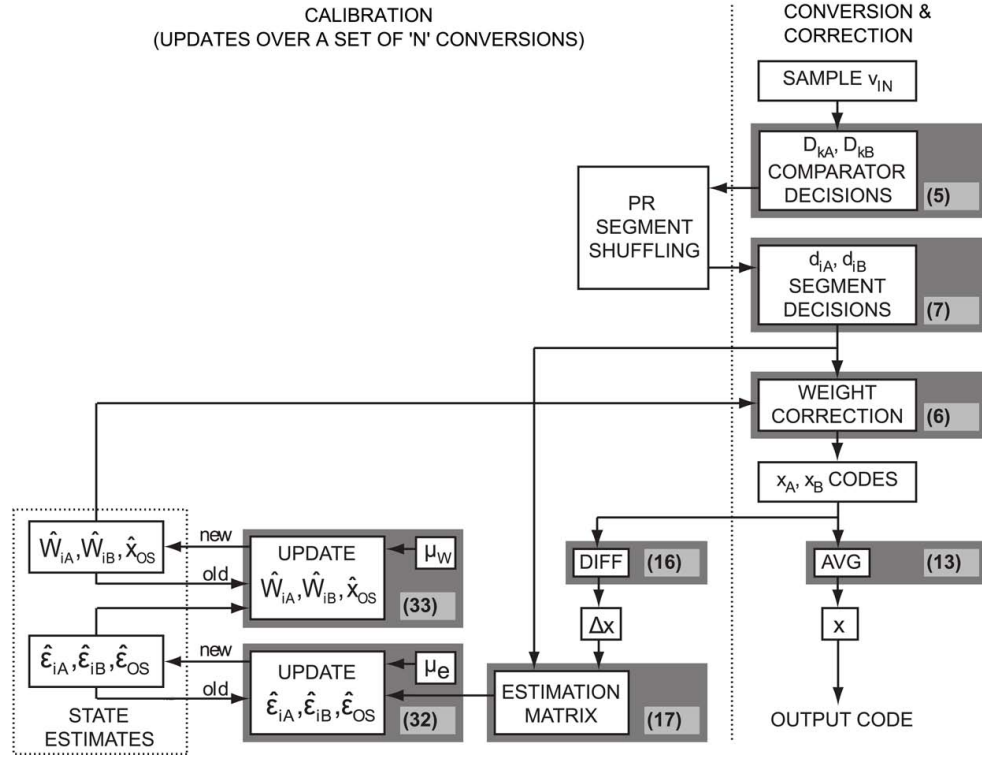


Fig. 5. Split SAR ADC Calibration, correction flowchart

timization matrix. Using the iterative solution method of [4], only the diagonal entries are needed to provide approximate values of ε_{iA} , ε_{iB} , which are continuously updated zero-bias estimates of the error in the estimated weights $\hat{W}_{iA}$, $\hat{W}_{iB}$. Estimates $\hat{W}_{iA}$, and $\hat{W}_{iB}$ are updated in loops with LMS coefficients $\mu_e = 2^{-4}$ and $\mu_W = 2^{-13}$, which were chosen to give an adaptation time constant of approximately 300 000 conversions. This corresponds to 300ms at a 1MSPS conversion rate; depending on initial error, convergence at startup is typically completed within a few seconds.

The following sections provide more detail on several aspects of the correction and calibration algorithm shown in Fig. 5.

A. Digital Error Correction

In (6), the weights W_i are numerical quantities representing the mismatch in the analog domain of the capacitor values C_i . Note that (6) is a relationship in the analog domain derived from charge conservation and is always true to within the quantization error in (5). If we have estimates $\hat{W}_i$ in the digital domain of the analog W_i , from (6) we can calculate a digitally corrected estimate $\hat{x}$ of the “true” output code x from the segment decisions d_i by

$$\hat{x} = \sum_{i=1}^n \hat{W}_i d_i \quad (8)$$

Defining the error in the estimated weights $\hat{W}_i$ relative to the true analog weights W_i by

$$\hat{W}_i = W_i + \varepsilon_i \quad (9)$$

substituting into (8) gives

$$\hat{x} = \sum_{i=1}^n (W_i + \varepsilon_i) d_i = \underbrace{\sum_{i=1}^n W_i d_i}_{\text{CORRECT } x} + \underbrace{\sum_{i=1}^n \varepsilon_i d_i}_{\text{ERROR}} \quad (10)$$

From (10) we see that digital correction can be performed if the errors ε_i are sufficiently small. Thus the task for the digital calibration algorithm is to determine the estimated weights $\hat{W}_i$ with sufficient accuracy. The following section describes the background calibration algorithm which estimates the errors ε_i and implements an adaptive LMS loop to drive the errors to zero.

B. Split-ADC Error Estimation

When the die area of the SAR ADC shown in Fig. 1 is split into two independent halves as shown in Fig. 2, each half produces an independent output at the original resolution. The outputs are obtained from the comparator decisions using independent correction tables for the corresponding decision weights:

$$\hat{x}_A = \sum_{i=1}^n \hat{W}_{iA} d_{iA} \quad (11)$$

$$\hat{x}_B = \sum_{i=1}^n \hat{W}_{iB} d_{iB} \quad (12)$$

The converter output x is obtained by averaging the independent $\hat{x}_A$ and $\hat{x}_B$ outputs:

$$\hat{x} = \frac{\hat{x}_A + \hat{x}_B}{2} \quad (13)$$

As described in [4], the total analog area and power are essentially unchanged. The total noise is also unchanged: assuming noise is kT/C limited, the noise of each output is degraded by 3dB since the capacitance on each side is half the original value, but the factor of 3dB is restored by the averaging operation at the output.

To see how the split ADC enables background estimation of the errors in the W_{iA} and W_{iB} correction coefficients, we expand the expressions in (11) and (12) as in (10)

$$\hat{x}_A = \underbrace{\sum_{i=1}^n W_{iA} d_{iA}}_{\text{CORRECT } x} + \underbrace{\sum_{i=1}^n \varepsilon_{iA} d_{iA}}_{\text{"A" ERROR}} \quad (14)$$

$$\hat{x}_B = \underbrace{\sum_{i=1}^n W_{iB} d_{iB}}_{\text{CORRECT } x} + \underbrace{\sum_{i=1}^n \varepsilon_{iB} d_{iB}}_{\text{"B" ERROR}} \quad (15)$$

The expressions for the output codes $\hat{x}_A$ and $\hat{x}_B$ consist of two terms. Comparing with (6) shows that the first term, indicated with "CORRECT x ", corresponds to the correct output code. Since both ADCs are converting the same input, these must be equal to within quantization error even if the SAR decision sequences D_{iA} and D_{iB} and resulting segment decisions d_{iA} and d_{iB} are different. The second term corresponds to the errors in the "A" and "B" output codes. Note that if the errors ε_{iA} and ε_{iB} are driven to zero, then $\hat{x}_A$ and $\hat{x}_B$ will approach the "CORRECT x " value and the averaging operation in (13) will produce the correct output code.

In the split ADC approach, the difference Δx between the "A" and "B" outputs is used to provide information to the calibration loop [4], [5]. Taking the difference we have

$$\Delta x = \hat{x}_B - \hat{x}_A = \underbrace{\sum_{i=1}^n \varepsilon_{iB} d_{iB}}_{\text{"B" ERROR}} - \underbrace{\sum_{i=1}^n \varepsilon_{iA} d_{iA}}_{\text{"A" ERROR}} \quad (16)$$

Thus the difference Δx depends only on the estimate errors ε_{iA} and ε_{iB} , with the dependency given by the decisions d_{iA} and d_{iB} , which are known. Since the subtraction in (16) cancels the input-dependent "CORRECT x " term, the input signal is removed from the calibration signal path. As discussed in [4], [5], this yields improved speed of calibration convergence relative to background calibration techniques which rely on decorrelation to remove input-dependent information from the calibration signal path.

The information in (16) describes how the errors ε_{iA} and ε_{iB} influence the difference Δx for one conversion. To estimate errors ε_{iA} and ε_{iB} , information from multiple conversions is necessary. Equation (17) shows a matrix representation of results from N conversions. Column vector Δ contains the Δx values from an "ensemble" of N conversions. Each row of the $\mathbf{R}$ matrix contains the corresponding segment decisions for

the "A" and "B" sides of each conversion. In the representation of (17) there are K segment weights to be calibrated on each side, contained in column vector $\mathbf{e}$, for a total of $2K$ errors to be estimated.

C. Accounting for Gain, Offset Errors

The previous analysis does not take into account errors due to gain or offset mismatch in each of the A and B channels. The fundamental principle of the split ADC approach is that Δx is zero when both ADCs are correctly calibrated and that any non-zero Δx is an indication that ADC linearity needs to be corrected. The problem is that any gain or offset mismatch errors between channels A and B will result in a non-zero Δx , even if the ADC linearity is correctly calibrated. In such a case, trying to drive the Δx to zero will corrupt the linearity calibration procedure. To fix this problem, we devise a technique to make the ADC calibration *insensitive* to the presence of mismatch errors.

We first modify (14) and (15) to model the effect of gain and offset error on the output codes .

$$\hat{x}_A = (1 + g_A) x + x_{OSA} + \sum_{i=1}^n \varepsilon_{iA} d_{iA} \quad (18)$$

$$\hat{x}_B = (1 + g_B) x + x_{OSB} + \sum_{i=1}^n \varepsilon_{iB} d_{iB} \quad (19)$$

in which g_A , g_B and x_{OSA} , x_{OSB} represent output-referred scale-factor and offset errors. Taking the average of the equations in (18) and (19):

$$\frac{\hat{x}_A + \hat{x}_B}{2} = x \left(1 + \frac{g_A + g_B}{2} \right) + \frac{x_{OSA} + x_{OSB}}{2} + \frac{1}{2} \sum_{i=1}^n \varepsilon_{iA} d_{iA} + \frac{1}{2} \sum_{i=1}^n \varepsilon_{iB} d_{iB} \quad (20)$$

Since we are calibrating only for ADC nonlinearity, and do not intend to correct offset and scale-factor errors in the overall ADC output code, we define x' including the effect of offset and linear scale-factor errors as

$$x' = x \left(1 + \frac{g_A + g_B}{2} \right) + \frac{x_{OSA} + x_{OSB}}{2} \quad (21)$$

Defining mismatch variables

$$g = g_B - g_A \quad (22)$$

$$x_{OS} = x_{OSB} - x_{OSA} \quad (23)$$

We can use (22) and (23) in (18) and (19) to express $\hat{x}_A$ and $\hat{x}_B$ as

$$\hat{x}_A = x' - \frac{g}{2} x - \frac{x_{OS}}{2} + \sum_{i=1}^n \varepsilon_{iA} d_{iA} \quad (24)$$

$$\hat{x}_B = x' + \frac{g}{2} x + \frac{x_{OS}}{2} + \sum_{i=1}^n \varepsilon_{iB} d_{iB} \quad (25)$$

Subtracting to get the difference Δx , the result is

$$\begin{array}{c} \Delta \\ \downarrow \\ N \text{ CONVERSIONS} \end{array} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \vdots \\ \Delta x_N \end{bmatrix} = \begin{array}{c} \mathbf{R} \\ \downarrow \\ 2K \text{ ERRORS} \end{array} \begin{bmatrix} d_{B(1,1)} & \cdots & d_{B(1,K)} & -d_{A(1,1)} & \cdots & -d_{A(1,K)} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ d_{B(N,1)} & \cdots & d_{B(N,K)} & -d_{A(N,1)} & \cdots & -d_{A(N,K)} \end{bmatrix} \begin{array}{c} \mathbf{e} \\ \downarrow \\ 2K \text{ ERRORS} \end{array} \begin{bmatrix} \epsilon_{1B} \\ \vdots \\ \epsilon_{KB} \\ \epsilon_{1A} \\ \vdots \\ \epsilon_{KA} \end{bmatrix} \quad (17)$$

$$\Delta x = x_B - x_A = \sum_{i=1}^n \epsilon_{iB} d_{iB} - \sum_{i=1}^n \epsilon_{iA} d_{iA} + x_{OS} + gx \quad (26)$$

Estimation of gain error g can be incorporated into the algorithm for estimating the weight errors by constraining the sum $\sum W_{iA} + \sum W_{iB}$ to be constant. Then, the effect of any non-zero gain error will be represented as $\sum W_{iA}$ and $\sum W_{iB}$ being unequal.

If we had an estimate of the offset error $\hat{x}_{OS}$, then we would correct x_A and x_B as follows:

$$x_A = x' - \frac{x_{OS}}{2} - \sum_{i=1}^n \epsilon_{iA} d_{iA} + \frac{\hat{x}_{OS}}{2} \quad (27)$$

$$x_B = x' + \frac{x_{OS}}{2} - \sum_{i=1}^n \epsilon_{iB} d_{iB} - \frac{\hat{x}_{OS}}{2} \quad (28)$$

where $\hat{x}_{OS}$ is the estimated offset error. Defining the error in this estimate as

$$\epsilon_{OS} = \hat{x}_{OS} - x_{OS} \quad (29)$$

Now, taking the difference between x_A and x_B , we get:

$$\Delta x = x_B - x_A = [-1] \epsilon_{OS} + \sum_{i=1}^n \epsilon_{iB} d_{iB} - \sum_{i=1}^n \epsilon_{iA} d_{iA} \quad (30)$$

With the result of (30), we modify our estimation matrix from (17) to include the effect of offset error ϵ_{OS} as shown in (31). Offset mismatch information is incorporated into the matrix by the additional column of [-1] entries and the ϵ_{OS} element.

D. Accumulation of N conversions

The matrix (31) represents the relationship between Δx values from N different conversions and the associated $ADCA$ and $ADCB$ estimate errors ϵ_{iA} , ϵ_{iB} and ϵ_{OS} . Each conversion gives another value of Δx providing information on estimate errors in accordance with (32). Over a set of N conversions, we can construct an error estimation matrix as shown in (31). In the 16-bit SAR converter designed, the DAC has 5 capacitor banks each comprised of 16 individually assignable unit capacitors. Based on the segment assignment logic discussed in (III-B), comparator decisions D_i are assigned to unit capacitors as d_i . In principle, the matrix in (31) could be solved by inversion. However, this would be computation-intensive and require more complex digital hardware. Additionally, the problem of singularity in the estimation matrix might result in no information about error estimates ϵ_{iA} and ϵ_{iB} being obtained for some cases. To make the matrix non-singular, we

employ a segmentation and shuffling technique as described in IV-E. To simplify the digital hardware, we use an iterative technique as described in IV-F.

E. Need for Segmentation and Shuffling

As mentioned in [4] and [5], one of the desired criteria for CMOS ADC design is background calibration that does not interrupt the processing of the ADC input signal. Previous calibration techniques described in [11] and [13] are not independent of the ADC input signal. For example, when a DC input is applied, these calibration schemes do not have enough information to calibrate out errors.

In this work, the signal-independence of the calibration technique is achieved as shown in Figure 4. This segmentation and shuffling scheme makes the decision trajectories pseudo-random in nature, thus allowing enough variety in the trajectories for calibration information to be extracted. According to procedure detailed in IV-F, an iterative matrix technique is used to solve for error estimates. To avoid singularity in the estimation matrix, groups of capacitor segments are assigned to each comparator decision as directed by a pseudorandom "base" pointer. Additionally, in any given capacitor unit, and for any given conversion cycle, there is always one capacitor segment that is unused. This dynamic assignment of capacitor segments to corresponding comparator decisions, combined with the universal presence of an "extra" capacitor, ensures enough variety in the segment assignment choices to handle not only DC inputs, but also DC inputs near the full-scale range.

F. Solving by the Iterative Technique

An exact solution of (31) would require matrix inversion, which is both computationally intensive and prone to numerical problems. An exact solution is also unnecessary, since the LMS method does not need an accurate error estimate. Therefore we adopt the approximate matrix solution of [5] which applies an LMS iteration to the error estimation:

$$\begin{aligned} \epsilon_{iA}^{(new)} &= (1 - \mu_e) \epsilon_{iA}^{(old)} + \mu_e \left([\text{sgn}(\mathbf{R})]^T \Delta \right) \\ \epsilon_{iB}^{(new)} &= (1 - \mu_e) \epsilon_{iB}^{(old)} + \mu_e \left([\text{sgn}(\mathbf{R})]^T \Delta \right) \end{aligned} \quad (32)$$

Parameter μ_e affects the dynamics of the LMS loop convergence.

The estimated digital values $\hat{W}_{iA}$, $\hat{W}_{iB}$, and $\hat{x}_{OS}$ used for correction are updated continuously in the background by an

$$\begin{array}{c} \Delta \\ \downarrow \\ N \text{ CONVERSIONS} \end{array} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \vdots \\ \Delta x_N \end{bmatrix} = \begin{array}{c} \mathbf{R} \\ \left[\begin{array}{ccc|ccc|c} d_{B(1,1)} & \cdots & d_{B(1,K)} & -d_{A(1,1)} & \cdots & -d_{A(1,K)} & -1 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ d_{B(N,1)} & \cdots & d_{B(N,K)} & -d_{A(N,1)} & \cdots & -d_{A(N,K)} & -1 \end{array} \right] \end{array} \begin{array}{c} \mathbf{e} \\ \left[\begin{array}{c} \epsilon_{1B} \\ \vdots \\ \epsilon_{KB} \\ \epsilon_{1A} \\ \vdots \\ \epsilon_{KA} \\ \epsilon_{OS} \end{array} \right] \end{array} \begin{array}{c} \downarrow \\ 2K + 1 \text{ ERRORS} \end{array} \quad (31)$$

LMS feedback loop.

$$\begin{aligned} \hat{W}_{iA}^{(new)} &= \hat{W}_{iA}^{(old)} - \mu_W \hat{\epsilon}_{iA} \\ \hat{W}_{iB}^{(new)} &= \hat{W}_{iB}^{(old)} - \mu_W \hat{\epsilon}_{iB} \\ \hat{x}_{OS}^{(new)} &= \hat{x}_{OS}^{(old)} - \mu_W \hat{\epsilon}_{OS} \end{aligned} \quad (33)$$

The LMS loop uses $\hat{\epsilon}_{OS}$, $\hat{\epsilon}_{iA}$, and $\hat{\epsilon}_{iB}$, which are estimated errors in the $\hat{x}_{OS}$, $\hat{W}_{iA}$, and $\hat{W}_{iB}$ values. The LMS multiplier μ_W controls the dynamics of the loop convergence. A key advantage of the LMS approach is that the $\hat{\epsilon}_{OS}$, $\hat{\epsilon}_{iA}$, and $\hat{\epsilon}_{iB}$ need not be accurate; all that is required is that they be zero-bias and (on average) steer the convergence of (33) in the correct direction. The general LMS technique is well established [9], [11], [12]; the novel contributions of this work are in the use of the "Split ADC" approach to extract calibration information in the background.

By constraining $\sum W_{iA} + \sum W_{iB} = \text{constant}$, we incorporate gain error calibration into the calibration of the decision weights. The effect of applying this constraint is that any non-zero gain error in the ADC results in $\sum W_{iA}$ and $\sum W_{iB}$ being unequal. In this work, the constraint is applied in an iterative fashion by keeping track of the change in weight sums $\sum W_{iA}$ and $\sum W_{iB}$ as shown in (34). This change in the sum of the weights is expressed as a scalar ratio h as shown in (35). This scalar is then used to modify the weights $\hat{W}_{iA}^{(new)}$ and $\hat{W}_{iB}^{(new)}$ in (33) to obtain new weights $\hat{W}_{iA}^{(new)'}$ as shown in (36).

$$\begin{aligned} \sum W_{iA}^{(old)} + \sum W_{iB}^{(old)} &= A^{(old)} \\ \sum W_{iA}^{(new)} + \sum W_{iB}^{(new)} &= A^{(new)} \end{aligned} \quad (34)$$

$$\frac{A^{(old)}}{A^{(new)}} = h \quad (35)$$

$$\begin{aligned} W_{iA}^{(new)'} &= h \cdot W_{iA}^{(new)} \\ W_{iB}^{(new)'} &= h \cdot W_{iB}^{(new)} \end{aligned} \quad (36)$$

Applying this constraint on the weights forces the gain error of the ADC to manifest only as $\sum W_{iA}$ and $\sum W_{iB}$ being unequal. The implementation of this constraint is represented in the overall algorithm as shown in Fig 5. It involves an additional update of the state estimates $\hat{W}_{iA}^{(new)}$ and $\hat{W}_{iB}^{(new)}$ based on the current value of scalar h .

Table 1. Behavioral Simulation Parameters.

PARAMETER		VALUE
ADC	Resolution	16b
	Conversion Speed	1 MSps
Comparator	SAR Decisions	20
	Input Referred Noise	-96 dBFS
DAC Capacitance	Total	16pF
	Minimum Unit	50 fF
	Input Referred Mismatch	50ppm
LMS Parameter	μ_W	2^{-13}
	μ_e	2^{-4}
Internal Digital Precision		24b

V. RESULTS

A 16-bit 1 Msps SAR converter was designed in a $0.25\mu\text{m}$ process, with capacitive DAC segmentation and shuffling to implement the algorithm described above [10]. The layout was extracted to a combination SPICE and HDL model to be tested with a MATLAB simulation of the calibration algorithm. Simulation parameters are shown in Table 1.

Figures 6a and 6b show plots of DNL and INL (in LSB) before calibration. Once the LMS calibration loop has converged to its steady state, the simulated linearity is better than $\pm 1\text{LSB}$. Figure 7 shows ADC error in dB relative to full scale (dBFS) plotted as a function of conversion for different input signals: random, sinusoidal, and DC equal to 0.9 of full scale (FS). In all cases, performance within 200 000 conversions. Figure 8 show the convergence of the maximum estimation parameter error for different input signals. In all cases, the error is below the limit required for proper ADC operation within 200 000 conversions. In similar fashion to [5], the ultimate limit on estimation parameter error is determined by either the internal digital precision or the value of the LMS parameters.

VI. CONCLUSION

A successive approximation ADC using the split ADC architecture has been presented. DNL and INL errors due to capacitor mismatch are digitally corrected and calibration is performed continuously in the background with a fully digital implementation. Behavioral simulation of a 16b 1Msps ADC shows calibration convergence within 200 000 samples.

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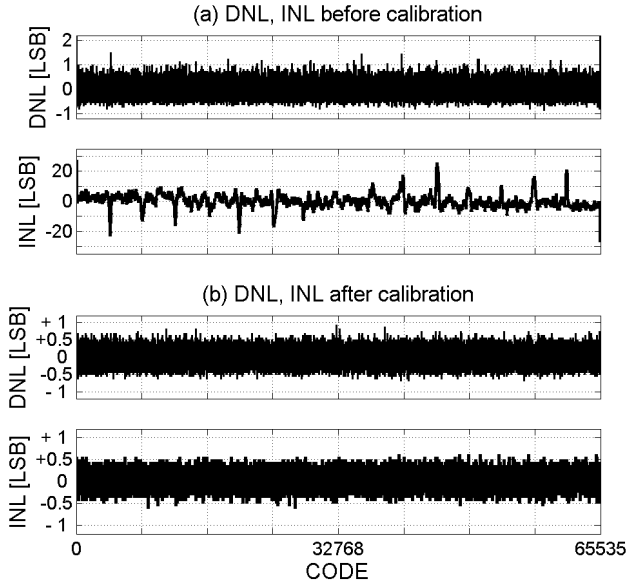


Fig. 6. (a) DNL and INL before calibration (b) DNL and INL after calibration.

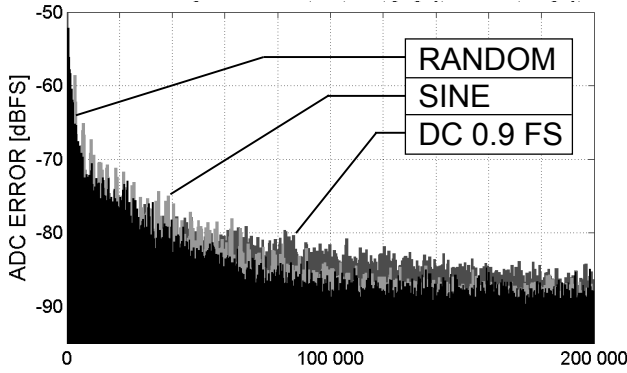


Fig. 7. ADC error convergence for different inputs

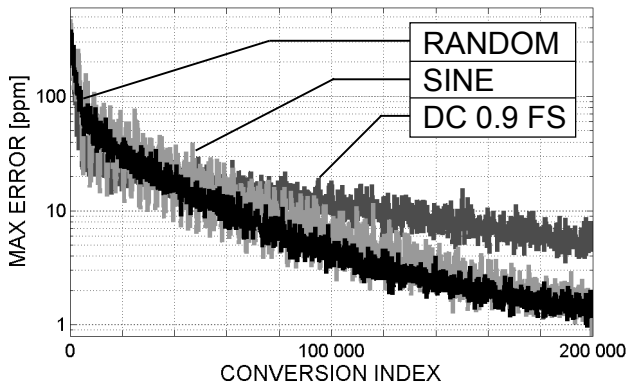


Fig. 8. Estimation parameter error convergence for different inputs

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